

TaSain Thomas

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SUMMARY

AI/ML Engineer focused on turning published research into working, production-grade systems. Built an uncertainty-aware BMI estimator in PyTorch. An SE-DenseNet121 trained on 7,208 real-world images, extended with split conformal prediction (90.4% verified coverage) and an LLM-based reasoning layer. Reaching 2.81 BMI-point MAE (11.56% MAPE). Also productionized MLOps infrastructure: a Kubernetes inference cluster with GPU passthrough serving an LLM. Industry 4.0 background at a major automotive manufacturer, where vision-guided assembly and a custom-worker-matching engine were adopted by the plant. Coursework in AI safety: LoRA/QLoRA, RLHF, red teaming, machine unlearning. Seeking an ML/AI engineering role where research translation and rigorous validation matter.

EDUCATION

Wayne State University, James and Patricia Anderson College of Engineering

Bachelor of Science, Computer Science

May 2026

Detroit, MI

• **GPA:** 3.4/4.0

• **Coursework:** LoRA, QLoRA, RLHF, DPO, GRPO, Red Teaming, Machine Unlearning, Grounded VLMs, Agentic AI Safety, Chain-of-Thought Reasoning, HarmBench, SafeChain

CORE COMPETENCIES

CORE COMPETENCIES: Research-to-production translation, Uncertainty quantification & conformal prediction, AI agent orchestration & prompt engineering, Model validation, Technical leadership

TECHNICAL SKILLS: Python, C++, SQL, SQLite, JavaScript, Dart, PyTorch, Conformal Prediction, Uncertainty quantification, LLM-as-Judge, GPT-4.1-Nano, Docker, Proxmox, GPU Passthrough, DigitalOcean, Nginx, systemd, Stripe, Telegram Bot API, Git/GitHub Workflow, Inductive Automation Ignition, Vuforia Model Targets

PROFESSIONAL EXPERIENCE

Industry 4.0 / Automation Intern (Rotational)

Sep 2023 - Sep 2024

Magna Electric Vehicle Structures

St. Clair, MI

- Built a worker-to-station assignment engine (C++/SQLite) with an evaluateFit() scoring algorithm balancing line familiarity, station expertise, and rank - adopted by the plant and used to assign workers during real shift changes.
- Developed an AR-based vision inspection system for tablet using Vuforia Model Targets and Inductive Automation Ignition: scanned parts, verified screw placement, and logged pass/fail criteria - fed the system with CSV data pipelines and Python API scripts.
- Built managerial monitoring systems tracking daily production output, employee attendance, and custom filtering, with role-based authorization (supervisor vs manager tiers) where manager-level write access to the test database was logged for auditing.

ENGINEERING PROJECTS

ConformalBMI: Uncertainty-Aware Body Mass Estimation from Images [GitHub](#)

Apr 2026 - Present

- Translated published BMI-estimation research (Digital Scale; Sui et al.; Manichand et al.; Jin et al.) into a working SE-DenseNet121 (PyTorch) estimator, trained on Celeb-FBI (7,208 real images, 6,174 labeled). Reaching 2.81 BMI-point MAE (11.56% MAPE) on a held-out test set.
- Extended the model with split conformal prediction for distribution-free uncertainty. 90% prediction intervals ($q = 6.44$) with empirically verified 90.4% coverage on unseen samples.
- Layered a GPT 4.1-Nano LLM-as-judge reasoning module that explains each estimate in plain English and flags high-uncertainty and extreme-BMI tail cases.

Homelab - Privacy-First AI Inference

Mar 2026 - Present

- Built a self-hosted homelab for privacy-focused AI serving: Proxmox hypervisor with GPU passthrough, running an AI agent inside a dedicated virtual machine.
- Configured memory and resource allocation across nodes; assembled and connected the hardware end to end.

On-Wheels Detailing - Founder & AI Agent Orchestration [GitHub](#)

Mar 2026 - Present

- Founded and operate a detailing business with actual paying customers; wired up Stripe payment processing end to end.
- Orchestrated a self-hosted AI agent (on my homelab) to design and generate the business website, using prompt-engineered role-play. The agent assumed specialist roles (design, security, database management) and self-checked its output against them. With a privacy constraint that no customer data is persisted.
- Built a Telegram agent layer delivering daily morning briefings and real-time lead alerts.
- Automated scheduled social content curation (M/W/F) with hashtag and engagement targeting.

Client Project - PDF Editing & File Management (Flutter)

Jan 2026 - May 2026

Wayne State University

- Team lead of a 4-person team building a mobile collaboration app for a real client, layered on top of an existing product (flow charts, venn diagrams, accounting tools).
- Personally built the PDF file-management feature: an annotation layer that overlaid text on PDFs and let added changes be saved, edited, or deleted.
- Ran the team process end to end - three scheduled weekly meetings, client communication, deadline adherence - and proposed the GitHub workflow (branching, pull requests, issue tracking).